

Using Body Tracking for Involving Museum Visitors in Digital Storytelling

Çağlar Genç
caglar.genc@ulapland.fi
University of Lapland
Rovaniemi, Finland

Jonna Häkkinen
jonna.hakkila@ulapland.fi
University of Lapland
Rovaniemi, Finland

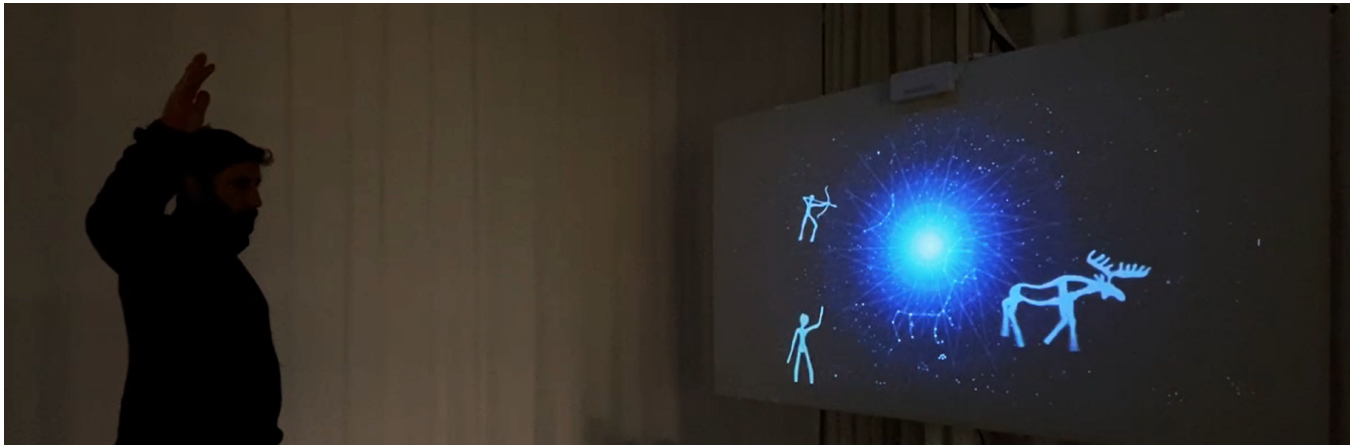


Figure 1: A user controlling the digital avatar in the Heavenly Hunt installation.

ABSTRACT

Technology involvement in museums leads to novel ways of communicating the museum content. Among other technologies, body tracking opens up the possibilities for embodied involvement in narratives presented in the museums. In this demo, we explore how body tracking could enable museum visitors to participate in historical storytelling installations. For this, we implemented a setup where the visitor's body is tracked via a motion capture device and projected as a character in the narration of an indigenous Sámi myth about the constellation of the stars, the Heavenly Hunt.

CCS CONCEPTS

• **Human-centered computing** → **Gestural input**; *Collaborative and social computing theory, concepts and paradigms*.

KEYWORDS

Motion tracking, museum, avatar, projection, installation, indigenous cultural heritage

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1 INTRODUCTION

Museums communicating historical and cultural stories have started to involve more and more interactive approaches. Interactive museum exhibitions have been reported to catch the interest of the audience better. The content use is more intensive if the presentation can be personalized according to the visitor's actions instead of predefined interactive content [7]. To complement the earlier introduced interactive public displays, tangible user interfaces [6], augmented reality (AR) and virtual reality (VR) are some popular technologies used in museum exhibitions [1]. In addition, motion tracking technology has gained attention. The bodily involvement of the visitors with museum installations is argued to provide more fun and engagement [2].

Full-body technologies open up an opportunity for visitors to take part in the digital content presentation, explored in prior works. Mishra and Cafaro examined body capture for exploring geospatial representation of data projected on a globe [9], and Kelly et al. implemented an installation where the young museum visitors can go through the metamorphosis of a butterfly by using full-body interaction [8]. Body RemiXer [3] is an installation that aims to create a social exchange environment between the visitors of the installation. This installation creates illustrations of the visitors' body (silhouettes) projected on the walls and shown through VR headsets.

In this demonstration, we focused on body tracking to investigate how to enable museum visitors to participate in a historical

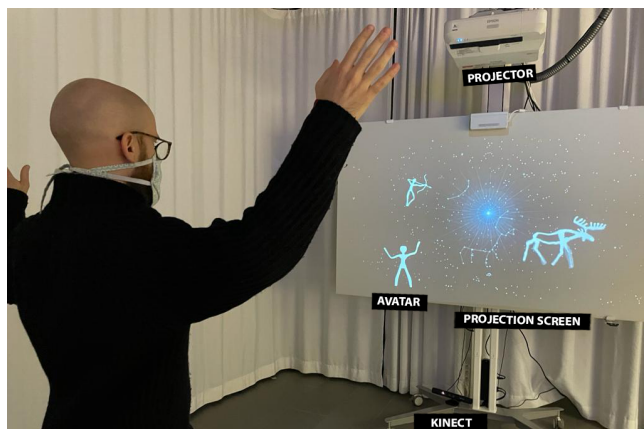


Figure 2: Implementation setup.

storytelling installation. We place our design in the context of designing with indigenous cultural heritage, specifically, the Sámi people of northern Scandinavia. The Sámi culture has been heavily influenced by the colonial history and assimilation policies [4], which makes the design context culturally sensitive. The cultural heritage content is highly important for the Sámi community and requires a respectful approach from the designers both in the design process as well as in handling the cultural heritage materials [10]. Today, there is increasing attention devoted to the revitalisation of the culture, with the objective of passing the stories and knowledge to the younger generations. Here, digital technologies offer new possibilities to learn and engage with the stories and traditions.

2 DESIGN & IMPLEMENTATION

Our aim is to involve the visitors in historical storytelling in museums. For this, we focused on the story of the Heavenly Hunt and implemented an installation in which the visitors can take part in this story by controlling a digital character with their bodies.

2.1 The Heavenly Hunt

Star installations in the sky are of origin for many indigenous myths. One of these is a narrative of Heavenly Hunt. The Heavenly Hunt is a traditional Sami myth, written down by Anders Fjellner, a priest, poet, and collector of Sámi folktales in the 19th century. Heavenly Hunt narrates a story of a reindeer hunt in which the shape of the stars resembles the mythological characters of Sami history. This story is turned into a video narration by the South Sami Museum and Cultural Center [12]. The video animates the hunting story by representing the mythological characters with symbolic images appearing over the stars on the sky. Also, the video is supplemented with audio narration by Kjersti Kweli.

2.2 Implementation

Our implementation aimed at involving the visitors in the South Sami Museum and Cultural Center's video narration of the Heavenly Hunt [12] by using their bodies. For this, we used a motion capture camera to detect the body movement of the visitor and mirrored the body data real-time to a digital avatar in the same style

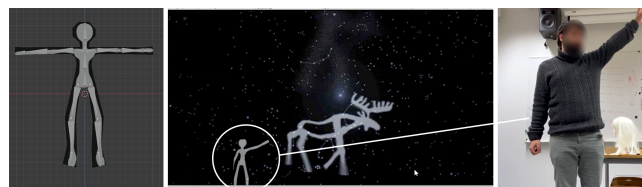


Figure 3: Body mapping of the avatar. (Left) Rigged mesh in Blender, (middle) Unity interface while the user moves, (right) user while controlling the avatar.

of the character illustrations that are used in the video (Figure 3). The avatar of the visitor is then embedded into the video narration as an audience in the story. When a visitor stands in front of the camera, the avatar appears among other characters and the visitor controls the movement of the avatar with her gestures and postures while watching the composition from a projected screen (Figure 1). Within the composition, the avatar of the visitor moves alongside the other illustrations, being a part of the story.

In our prototype (Figure 2), we capture the body movements of the visitors via Microsoft Kinect 360 Xbox. The real-time data coming from this device is rigged to a 2D mesh created via Blender software and overlaid onto the video narration by using Unity (Figure 3). This composition is, then, projected on a screen with a short-throw projector (Epson eb-1460ui).

3 DISCUSSION

HCI research has many aspects where it needs to consider cultural sensitivities when designing interactive applications [5]. Conducting research with indigenous communities requires respectful partnership [11], and this should be reflected also in the HCI design applied to the cultural heritage. In our research, we seek to find a design solution that combines the engaging experience of interactive technology, but with a style that is faithful and respects the indigenous cultural heritage design context. As future work, we seek to evaluate our prototype with a user study. Especially, we will explore the design aspects with respect to the cultural sensitivity, and develop the concept and prototype further with a co-design approach.

Our work opens also the floor to the discussion related to a new dimension in the augmented humans research theme. Can we augment humans to explore and reach back in history, in myths and other cultures? The design for augmented humans is not only for the present, but it could bring experiences that go beyond the now, and connects us with the past. Relating oneself with presentations that augment our understanding and empathy with history and heritage can be, in our belief, an interesting future research direction for the augmented humans community.

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